ELECTRICAL AND MAGNETIC FIELD CALCULATOR

Based on EPRI Transmission Line Reference Book (Red Book, 1982)

Version 1.3

Project/Site: *Effect of Clearance on EMF of typical 345kV double circuit line.*

Wire Data NW corner of house on Smith Rd

Number of Wires: 10

Wire Label	Description	Voltage (kV)	Current (amps)	Phase Angle (deg)	Bundle Spacing (in)	Conduc. Diam (in)	Number in Bundle	Location X (ft)	Location Y (ft)	Equivalent Diam (ft)	Equivalent Diam (in)
1		0	0	0	0	0.546	1	30	148.3	0.046	0.546
2		0	0	0	0	0.546	1	10	148.3	0.046	0.546
3-1		345	2695	240	18	1.14	2	10	109.8	0.534	6.406
3-2		345	2695	0	18	1.14	2	10	46.1	0.534	6.406
3-3		345	2695	120	18	1.14	2	10	40.5	0.534	6.406
6-1		345	2260	0	0	1.762	1	-75.6	45.6	0.147	1.762
6-2		345	2260	120	0	1.762	1	-50	45.6	0.147	1.762
6-3		345	2260	240	0	1.762	1	-24.4	45.6	0.147	1.762
4-1		0	0	0	0	0.546	1	-68	73.2	0.046	0.546
5-1		0	0	0	0	0.546	1	-32	73.2	0.046	0.546
										0.000	0.000
										0.000	0.000
										0.000	0.000
										0.000	0.000
										0.000	0.000
										0.000	0.000
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										0.000	0.000
										0.000	0.000
										0.000	0.000

Use GREEN text to copy/paste with PLS-CADD EMF Calculator

X location along ground: 50 ft

Y location above ground: 3.28084 ft

Graphing

Left Bound: -125 ft

Center for Mag: 0 ft

Center for Elec: 0 ft

Right Round: 125 ft

Values at Given Location (X=50.00, Y=3.28)

Magnitude of Magnetic Field (App. 8-1)	**144.0**	mG
Magnitude of Mag. Field (Resultant)	**144.1**	mG
Magnitude of Electric Field (App. 8-1)	**0.647**	kV/m
Magnitude of Electric Field (Resultant)	**0.647**	kV/m

Maximums over Given Range

Max Mag Field (using Appendix 8.1)	198.58	mG
Approx. Location	-68.36	ft
Max Mag Field (using Resltnt of B_r & B_i)	**224.31**	mG
Approx. Location	**-68.36**	ft
Max Elec Field (App. 8-1)	2.345	kV/m
Approx. Location	12.61	ft
Max Elec Field (Resultant of E_r & E_i)	**2.347**	kV/m
Approx. Location	**12.61**	ft



